

A General Purpose, Broadband Transimpedance Amplifier

by

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ABSTRACT

A transimpedance amplifier is described which has been specially designed to work with a large range of diode capacitances and feedback resistor values at broad bandwidths. The amplifier achieves both low voltage and current noise and is DC coupled with millivolt offsets. The input stage includes a limiter to improve overdrive recovery time. The circuit is capable of 200MHz analog bandwidths at 10k Ω transimpedance and 10pf diode capacitance. It will be implemented in a complementary bipolar process (ATT cbicV2).

Transimpedance amplifiers working from capacitive sources such as photodiodes are difficult to implement with conventional voltage feedback op-amps. A photodiode transimpedance amplifier bears strong resemblance to a differentiator circuit, which requires compensation in the form of a feedback capacitor to ensure stability. Unfortunately, the addition of a feedback capacitor directly impacts the transimpedance bandwidth (see figure 1).

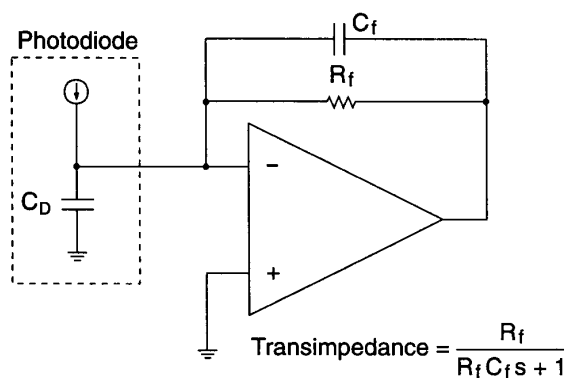


Figure 1

An op-amp with a capacitive inverting input impedance requires compensation because the capacitance at the inverting input introduces an additional loop transmission pole. The pole is at the time constant $R_f C_d$, where C_d is the capacitance at the inverting input (photodiode capacitance). Without compensation, the loop is potentially unstable due to the additional pole.

A better way to build a transimpedance amplifier is with low loop gain, using the $R_f C_d$ time constant as the dominant loop pole with no additional poles in the pass band of the amplifier. In this case, the amplifier must have a very broad bandwidth (open loop) compared to an op-amp, but only requires a moderate open loop gain. The integrated circuit discussed herein uses this approach. The concept of using a broadband, low open loop gain is not new [1,2], however, this implementation offers several areas of significantly improved performance, particularly at high diode capacitances.

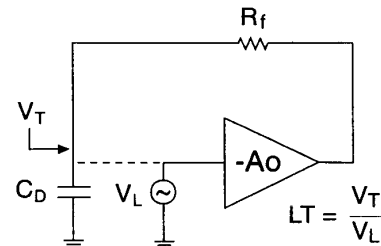


Figure 2

Analyzing the low gain transimpedance amplifier for closed loop bandwidth is best done from a loop transmission (LT) viewpoint. Breaking the feedback loop and computing the gain encountered traversing the loop (see figure 2):

$$LT = - \frac{A_o}{R_f C_d s + 1}$$

where A_o is the open loop gain of the amplifier. (Note that the A_o term is constant, with no dynamics. The open loop gain of the amplifier must have much broader bandwidth

than the closed loop bandwidth.) The closed loop bandwidth of the system is that frequency where the LT reaches unity, and for the above expression is at about:

$$\frac{A_o}{2\pi R_f C_d} \text{ Hertz}$$

What this suggests for a general purpose transimpedance amplifier is that A_o , the open loop gain, should be adjustable. Adjustable gain allows a single device to operate over a large range of diode capacitances and feedback resistances while still making possible a high closed loop bandwidth.

In this specific circuit, the open loop gain is adjustable over a range of $\times 10$ to $\times 100$, with an open loop bandwidth of 400MHz throughout this range. This makes possible closed loop bandwidths of 200MHz at a $10k\Omega$ transimpedance with a 10pf diode capacitance. To achieve this performance with an op-amp would require the use of an 80 femtofarad feedback capacitor, which is smaller than the parasitic associated with a leaded resistor. The op-amp would also require a 25GHz gain bandwidth product (as can be seen by taking the ratio of the capacitances times the bandwidth). There are special purpose transimpedance amplifiers available using this low open loop gain approach, but no general purpose, broadly adjustable transimpedance amplifiers are available.

The broad bandwidth in this circuit is achieved by using two gain stages (see figure 3). The inputs to the first stage are buffered with emitter followers (so that the input capacitance and input bias current are low). The first stage consists of a cascoded differential pair with resistive loading. The second stage is a differential voltage to current converter built out of a pair of closed loop buffers (to ensure low distortion with large input signal swings). The closed loop buffers also allow a 10:1 gain variation with only small changes in bandwidth. The second stage output current is converted to a single ended voltage by means of current mirrors driving a resistor. The resulting voltage is passed through a buffer to the output.

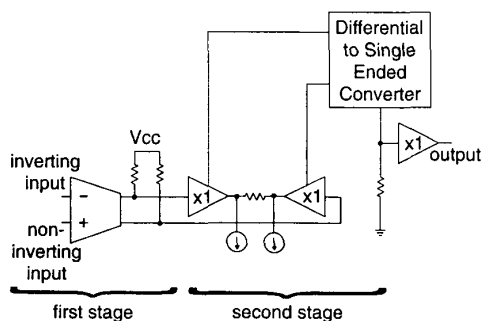


Figure 3

In any transimpedance amplifier, both current and voltage noise are important. Current noise contributions are

generally flat with frequency up to the closed loop bandwidth (ignoring $1/f$ noise terms). Voltage noise from the amplifier is differentiated due to the capacitive inverting input impedance. As frequency increases, the output noise due to input voltage noise increases, with output noise shape proportional to frequency. In a wide bandwidth transimpedance amplifier, the amplifier's input voltage noise can easily dominate the broadband noise measured at the amplifier's output.

Several things were done to minimize total output noise. The inverting input bias current needs to be low to reduce input current noise, but there is no corresponding requirement on the non-inverting input bias current.

The inputs of the first gain stage are buffered with emitter followers which are biased at different currents, with the non-inverting side operating at four times the current of the inverting side (see figure 4). The areas of the emitter followers are correspondingly mismatched, resulting in low offset voltage and correlated input bias currents (with a 4x magnitude difference). This is done to improve noise. If the two emitter followers are run at the same currents, the total voltage noise power is doubled over that for a single device. By running the non-inverting side at four times the current of the inverting side, the input voltage noise of the amplifier approaches that of the single inverting input transistor's noise, resulting in almost half the noise power of a conventional approach. By using this bias arrangement, the circuit achieves 3.0nV/Hz input voltage noise and 0.5pA/Hz inverting input current noise.

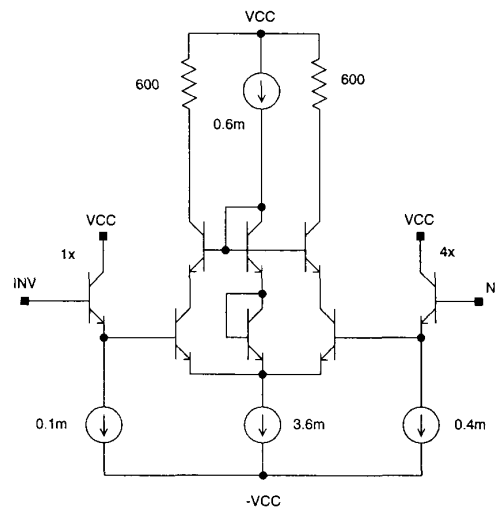


Figure 4

Another problem encountered in transimpedance amplifiers is that of overdrive recovery. It is fairly straightforward to build the amplifier so that it does not saturate internally as the result of an overdrive, but the feedback loop itself can

still saturate. If the output of the amplifier limits under overdrive conditions, the excess current from the diode can no longer flow through the feedback resistor and acts to charge the diode capacitance. When the overdrive condition stops and the diode current drops to a low value, the charge stored on the diode capacitance can only discharge through the feedback resistor, resulting in a long recovery time constant proportional to $R_f \cdot C_d$.

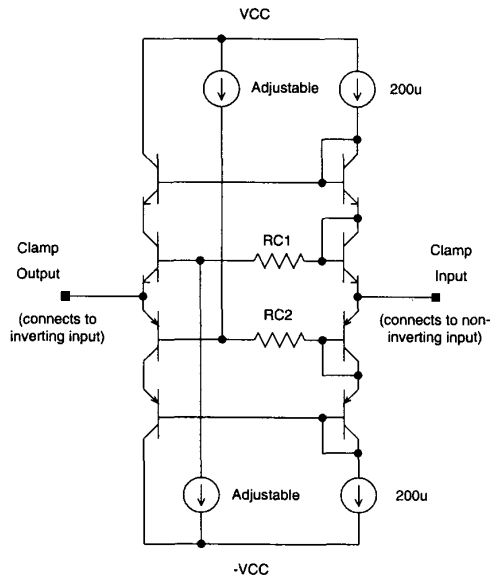


Figure 5

In order to deal with an overdrive in a more graceful manner, this circuit has an input limiter. A buffer is connected between the non-inverting input and the inverting input. The output side of the buffer is kept unbiased by means of the voltage drop across the resistors R_{c1} and R_{c2} (see figure 5). During an overdrive condition, when the inverting input wanders away from the non-inverting input by an amount comparable to the drop across R_{c1} or R_{c2} , the buffer output stage turns on, clamping the inverting input voltage. The input must be allowed some freedom of movement due to the low open loop gain of the amplifier. Therefore, the limit voltage is adjustable, allowing the user to trade off limiter induced harmonic distortion against overdrive recovery time.

A transimpedance circuit has been described which is versatile enough to cover a wide range of feedback resistances and source capacitances while maintaining broad closed loop bandwidth. The circuit also achieves low input current and voltage noise by means of special input stage design. A limiter circuit in the input stage allows rapid overdrive recovery.

REFERENCES

- [1] T. V. Muoi, "Receiver Design for High-Speed Optical-Fiber Systems", IEEE J. of Lightwave Technology, Vol. LT-2, No. 3, June 1984.
- [2] R. G. Meyer and R. A. Blauschild, "A Wide-Band Low-Noise Monolithic Transimpedance Amplifier", IEEE J. of Solid State Circuits, Vol. SC-21, No. 4, August 1986.